

Week 34: Probability theory

Some practical info:

- Interactive lectures means:
 - I will go over some relevant theory
 - You will have a bit of time to work on exercises
 - I will go over my suggestions for solutions
- The timeslot is terrible. Please suggest alternative times that would work for you?
- I am away throughout October, so no lectures then. Will inform on Wiki.

Sample spaces, events and measurable spaces

Theory

Informally, a *random experiment* has the following objects:

1. Sample space: Set Ω with possible outcomes $\omega \in \Omega$.
2. Events: Subsets of Ω for which probabilities are defined.
3. Measurable space: Sample space Ω equipped with a family of events $\mathcal{E}$.

Definition 1.1. (*σ -algebra*) A σ -algebra $\mathcal{E}$ on Ω is a family of subsets of Ω satisfying

- $\Omega \in \mathcal{E}$
- If $A \in \mathcal{E}$, then $A^C = \Omega \setminus A \in \mathcal{E}$.
- If $A_1, A_2, \dots \in \mathcal{E}$, then $\cup_{i=1}^{\infty} A_i \in \mathcal{E}$.

Definition 1.2. (*Events*) The family of events $\mathcal{E}$ is a σ -algebra on Ω .

Problem 1

Let $\Omega = \{1, 2, 3\}$ and $\mathcal{E} = \{\emptyset, \Omega, \{1\}, \{2, 3\}\}$.

1. Show that $(\Omega, \mathcal{E})$ is a measurable space.
2. Is $\{1, 2\}$ an event?
3. How many subsets of Ω are events?
4. Every event is a set, but every set is not an event. Explain this.

Probabilities and random variables

Theory

Definition 2.3. (*Probability*) A probability on $(\Omega, \mathcal{E})$ is a function $P : \mathcal{E} \rightarrow \mathbb{R}$ such that

- $P(A) \geq 0$ for all $A \in \mathcal{E}$.
- $P(\Omega) = 1$.
- If $A_1, A_2, \dots \in \mathcal{E}$ are pairwise disjoint, then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

We often call the tuple $(\Omega, \mathcal{E}, P)$ a *probability space*.

Definition 2.4. (*Random variable*) A random variable (RV) is a function $X : \Omega \rightarrow \mathbb{R}$ such that

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{E}$$

for every $B \in \mathcal{B}(\mathbb{R})$. Here $\mathcal{B}(\mathbb{R})$ denotes the Borel σ -algebra on $\mathbb{R}$, which is the smallest σ -algebra containing all open intervals in $\mathbb{R}$.

Proposition 2.5. *(Probability associated with RV) If X is a RV, then the function $P_X : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by*

$$P_X(B) := P(X \in B) = P(X^{-1}(B))$$

is a probability on $\mathbb{R}$. We will prove this in the last problem.

Definition 2.6. *(Cumulative distribution function associated with an RV) The cumulative distribution function (CDF) of X is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by*

$$F_X(x) := P_X((-\infty, x]) = P(X \leq x).$$

Problem 2

Let $X : \Omega \rightarrow \mathbb{R}$ be such that $(X \leq q)$ is an event for all $q \in \mathbb{Q}$.

1. Prove that $X^{-1}((-\infty, x]) = (X \leq x)$ is an event for every $x \in \mathbb{R}$.
2. Prove that $X^{-1}(I)$ is an event for every interval $I \subseteq \mathbb{R}$.
3. Define

$$\mathcal{E}_X := \{B \subseteq \mathbb{R} : X^{-1}(B) \in \mathcal{E}\},$$

and that $\mathcal{E}_X$ is a σ -algebra on $\mathbb{R}$.

4. Use the preceding parts to prove that X is a RV. Conclude that its CDF $F_X(x)$ is defined for every $x \in \mathbb{R}$.

Expectation

Theory

Definition 3.7. (*Simple RV*) A RV X is simple if it takes on only finitely many values. We can write this as $X(\Omega) = \{X(\omega) \mid \omega \in \Omega\} = \{x_1, \dots, x_n\}$ for a finite n . Defining $A_i = \{\omega \in \Omega : X(\omega) = x_i\}$, we have that $A_1, \dots, A_n$ is a disjoint partition of Ω , and X can be expressed as $X = \sum_{i=1}^n x_i \mathbb{1}_{A_i}$.

Definition 3.8. (*Expectation of simple RVs*) The expectation of a simple RV $X = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is defined as $\mathbb{E}[X] = \sum_{i=1}^n a_i P(A_i)$.

Problem 3

Let X and Y be simple RVs, and let $\alpha, \beta \in \mathbb{R}$. Prove

- Constants are preserved: If $X = c$ for some $c \in \mathbb{R}$, then $\mathbb{E}[X] = c$.
- Monotonicity: If $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.
- Linearity: $\mathbb{E}[\alpha X + \beta Y] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]$.
- Relation to probability: If $A \in \mathcal{E}$, then $\mathbb{E}[\mathbf{1}(A)] = P(A)$.

Problem 4

Let $X : \Omega \rightarrow \mathbb{R}$ be an RV, and let $\varphi : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ be a simple and measurable function.

1. Show that $P_X(B) = P(X \in B)$ defines a probability on $(\mathbb{R}, \mathcal{E}_X)$.
2. Show that $\mathbb{E}[\varphi(X)] = \mathbb{E}_X[\varphi]$ with $\mathbb{E}_X[\cdot]$ denoting the expectation with respect to the probability P_X .

To conclude...a more general definition of expectation